

「臺灣大學醫學院附設醫院 暨 臺北醫學大學」研究計畫申請書

Research Proposal for NTUH and TMU

一、基本資料

計畫編號		(承辦單位填寫)		
研究主題		☐ 幹細胞相關研究 ☒ 癌症免疫治療 ☐ 微創手術的應用 ☐ 胸腔醫學研究 ☐ 神經及精神醫學基礎臨床研究 ☒ 醫療及生物大數據分析 ☐ 免疫疾病的新興治療 ☐ 生醫材研究 ☐ 其他有助於醫療發展的主題		
總計畫 名稱	主持人	姓名：	職稱：	單位及部門：
	聯絡方式	聯絡電話：	E-mail：	
	中文	醛固酮在腎細胞癌發病和腫瘤形成角色		
	英文	Aldosterone and Its Role in the Pathogenesis and Progression of Renal Cell Carcinoma		
子計畫 名稱 (一)	主持人	姓名：	職稱：	單位及部門：
	聯絡方式	聯絡電話：	E-mail：	
	中文	原發性醛固酮病患者中癌症的盛行率和流行病學趨勢		
	英文	Unveiling Cancer Prevalence and Epidemiological Trends in Patients with Primary Aldosteronism		
子計畫 名稱 (二)	主持人	姓名：	職稱：	單位及部門：
	聯絡方式	聯絡電話：	E-mail：	
	中文	醛固酮與腎細胞癌之細胞模型路徑和機制探討		
	英文	Aldosterone and Renal Cell Carcinoma: Elucidating Pathways and Mechanisms Across Model Systems		
子計畫 名稱 () (請自行 增加)	主持人	姓名：黃俊仁	職稱：副院長	單位及部門：萬芳醫院麻醉科
	聯絡方式	聯絡電話:0970746010	E-mail：cjhuang@tmu.edu.tw	
	中文	醛固酮與轉移性腎細胞癌之關聯性的外泌體:多體學探勘		
	英文	Exosomes associated with aldosterone and metastatic renal cell carcinoma: multi-Omics approach		
計畫執行期限		113 年 1 月 1 日至 114 年 12 月 31 日		
研究性質		☒ 基礎研究 ☒ 臨床試驗 ☐ 轉譯醫學 ☐ 技術發展 ☐ 其他_____		
是否進行下列實驗		☒ 人體研究 ☒ 動物實驗 ☐ 基因重組試驗 (若有勾選需附相關實驗許可文件)		
本案聯絡人姓名：			電話：	E-mail：

☒主持人及研究團隊皆保證此所提之計畫內容與院內、院外其他計畫未重複，且未抄襲相關論文，願意負擔一切責任，並恪遵學術誠信原則。

計畫主持人簽章(含子計畫主持人)：_____

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二、總計畫/Main Project

繳交計畫前，請刪除說明/Please delete all the instructions before proposal submission.

計畫書全部內容限 15 頁為原則，總計畫及子計畫說明分別包含下列項目：

Total restricted to 15 pages for Main and Subprojects, and including the following sections:

1. 摘要與計畫願景/Synopsis and Long-Term Goals of the Project

2.

2. 計畫背景、領域現狀、關鍵問題、本計畫目的與臨床重要性/

Background, Current Status, Outstanding Problems, Objectives, and Clinical significance of the Project

This section presents issues addressed by the proposal.

Please describe the current state-of-the-art and ongoing developments in fields relevant to this proposal. Provide an assessment of what further breakthrough needs. Comment on the position of your proposal with respect to the context developed above: for example, the position of the projected results at national or international level. With respect to competing with or complementary to literature work, patents and standards will be another example. Please cite necessary references.

Primary Aldosteronism, the leading cause of secondary hypertension, is known to result in a heightened degree of target organ damage. Numerous epidemiological investigations conducted over the past twenty years indicate that the prevalence of PA exceeds 5%, and potentially even 10%, among hypertensive patients in both general and specialty healthcare contexts [1-2]. The defining characteristic of PA is the unregulated overproduction and release of aldosterone, resulting in hypertension and consequential target organ harm and malfunction, the severity of which is often more pronounced than that seen in other hypertensive patient groups [3-7].

Aldosterone, classically known for its regulatory effects on extracellular volume and blood pressure through epithelial actions, also exerts multiple biological actions in non-epithelial tissues [8]. In recent years, multiple human and experimental animal studies have affirmed the significant role of aldosterone in the initiation and progression of renal injury, inflammation, and subsequent fibrosis [9, 10]. These studies elucidate that aldosterone triggers the production of reactive oxygen species (ROS) [11] and induces cellular proliferation or deformation [12] in cultured mesangial cells.

Furthermore, aldosterone promotes collagen synthesis in renal fibroblasts in culture [13], and plays a part in the regulation of fibrosis, cellular growth, and cell death [14]. However, the role of aldosterone in renal cell growth and metastasis, as well as its potential influence on other types of cancers in the context of aldosteronism, remains largely unexplored [15]. Given that hypertension and obesity [16], both associated with increased aldosterone levels [17], are established risk factors for advanced renal cell carcinoma (RCC), aldosteronism patients provide a viable human model for studying aldosterone's effects on major organ injury and malignancy.

RCC represents the most prevalent form of kidney malignancy [18], with its incidence showing an upward trend in most countries over the past decade [19]. Ranking as the sixth and ninth most common cancer among men and women, respectively, in the United States [20], RCC also poses a significant health burden in Taiwan, where the age-standardized incidence rate in 2012 was 5.32 and 2.49 per 100,000 persons for men and women, respectively [Bureau of Health Promotion, 2015]. Almost 25% of all kidney cancer patients present with locally advanced or metastatic RCC, a

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condition with a poor prognosis [21]. Traditional therapies, such as chemotherapy and radiation therapy, are ineffective, with only 10 to 20% of patients benefiting from immunotherapy [22; 23]. Although recent advancements in understanding the etiology of advanced RCC have facilitated the development of angiogenesis inhibitor agents, metastatic RCC (mRCC) generally remains resistant to chemotherapy and hormonal therapy, with only marginal sensitivity to immunotherapy [24]. Despite the availability of several promising therapeutic strategies, nearly all patients succumb to the metastatic disease, prompting ongoing research to identify RCC-specific biomarkers for improving early diagnosis, monitoring tumor progression, and predicting patient prognosis [25].

Aldosterone has been reported to up-regulate p21 in various cell types, including cancer cells [26], vascular smooth muscle cells, renal epithelial cells [27], and induce renal senescence [28] in proximal tubular cells through the mineralocorticoid receptor (MR). Intriguingly, thiazide diuretics, known to stimulate aldosterone production, have been associated with an increased risk of RCC in some studies [17]. The proposed series of investigations aim to elucidate whether aldosterone might contribute to the development and metastatic progression of RCC, and whether aldosterone inhibition could potentially mitigate or eliminate metastasis.

Previous studies from a large cohort of mRCC patients identified mTOR pathway activation as an adverse prognostic factor. Rapamycin was discovered to inhibit mTOR and thereby induce G1 arrest and apoptosis [29]. Despite the limited success of temsirolimus and everolimus in treating advanced RCC, due to the development of progression at metastatic sites during treatment, the median progression-free survival of patients treated with these drugs has been reported at 3.8 and 4.9 to 11.2 months, respectively [30]. However, the use of mineralocorticoid receptor blockers has nearly tripled overall survival, raising the possibility of a potential cure for advanced RCC.

Currently, the available non-invasive treatments for primary aldosteronism fall short of satisfactory outcomes in clinical practice. The protein kinase B (PKB or AKT)/mammalian target of rapamycin (mTOR) signaling pathway plays a critical role in the tumorigenesis and metastasis of various human tumors, including renal cancer and adrenal carcinoma. However, the necessity of this pathway in the pathogenesis of RCC following aldosterone stimulation remains unexplored. To address this, we aim to investigate the activity of the PI3K/AKT/mTOR signaling pathway following aldosterone stimulation in kidneys from normal individuals, primary aldosteronism patients, and human RCC cells.

In line with these considerations, this Innovative TMU-NTUH Joint Research Program Application carries significant importance and bears scientific implications for the TAIPAI study group to lead in the field of aldosterone-related malignancies."

3. 研究方法與整合方式/Experimental and Project Structure

This section provides rationales for the methods and approaches that will be tailored to lift the crucial barriers and thus to reach the objectives defined above.

Please break the work package down into major tasks. Provide interdependencies and responsibilities within the subprojects and describe how previous experiences qualify the PIs for the task. Please also provide short profiles of the PIs and other key participating personnel who will be undertaking the work.

4. 子計畫之間之整合程度與臨床研究交流合作之效益/

The integration between subprojects and the Contribution to Clinical collaboration.

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Dr. 黃俊仁 will be the PI of Subproject 3. The Subproject 3 will also include a co-investigator, Dr. 張肇源, a PhD researcher in the Department of Medical Research, Wan Fang Hospital, and the Graduate Institute of Clinical Medicine, Taipei Medical University. Dr. 張肇源 has his expertise in high throughput exosome isolation techniques and the fabrication of engineered exosomes. Dr. 張肇源, through collaboration with Dr. 黃俊仁, is currently conducting a multi-year NSTC project (111-2314-B-038-127-MY2) to explore the therapeutic potentials of engineered exosomes using the immortalized murine macrophages RAW264.7 cells. Through their future collaboration, Dr. 黃俊仁 & Dr. 張肇源 will develop exosome-based novel diagnosis of disease progress and engineered exosome-based novel therapies against ARDS-induced lung injury and fibrosis in a multi-year NSTC project (112-2314-B-038-137-).

Dr.黃俊仁, Dr.張肇源, & Dr.關士傑 also collaborate on a regular basis. Dr.黃俊仁, Dr.張肇源, & Dr.關士傑 are currently conducting a project funded by TMU-NTUH Joint Research Program (111-TMU12), aim of which is to establish a novel diagnostic platform for early prediction of disease progression from kidney diseases transition. Exosomes will be isolated from liquid biopsy samples collected from various kidney disease animal experiments and clinical data from kidney transplant patients. Profiling of exosome cargo will be conducted using bioinformatics and multi-omics approaches. The information will then be combined with clinical records to identify novel diagnostic biomarkers.

5. 參考文獻/References

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23. Wu VC, Chueh SC, Chang HW, Lin LY, Liu KL, Lin YH, Ho YL, Lin WC, Wang SM, Huang KH *et al*: **Association of kidney function with residual hypertension after treatment of aldosterone-producing adenoma.** *Am J Kidney Dis* 2009, **54**(4):665-673.
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三、子計畫/Subprojects

子計畫一 Subproject 1	子計畫名稱	原發性醛固酮病患者中癌症的盛行率和流行病學趨勢	主持人 PI	
	Title of Project	Unveiling Cancer Prevalence and Epidemiological Trends in Patients with Primary Aldosteronism		

繳交計畫前，請刪除說明/Please delete all the instructions before proposal submission.

1. 本子計畫目的/Specific Objectives of this Subproject

Describe, for this subproject, the specific objectives and the significance, *e.g.*, prospects for academic excellence, originality and innovations in scientific research or technological development, and impact on the research field and international competitiveness.

Specific Aims 1.1: To construct and expand the knowledge on the malignant outcomes in Primary Aldosteronism (PA) patients.

Specific Aims 1.2: Integrating the clinical outcomes of Primary Aldosteronism and associated malignancies across multi-institutional datasets

2. 經費編列

項目名稱	使用目的	總價
人事費	兼任研究助理(碩士級) (6000 元/月*12 月*2 人)	144,000

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		小計： 元
業務耗材費		
		小計： 元
總計：新台幣 元		

3. 人力配置

類別	姓名	現職	在本計畫內擔任之工作性質、項目及範圍
子計畫主持人	TBD		研究構思、大數據研究障礙排除、跨計畫研究整合、研究統計數據分析、計畫結果撰寫
協同主持人			協助研究構思、協助研究障礙排除以及實驗室作業、結果統整分析
研究人員			實驗室作業、行政處理、經費報支、實驗結果彙整

4. 本子計畫對整體計畫的貢獻/Contribution of this Subproject to the Synergic Integration

This section re-defines the questions of the main project yet from the aspect of this subproject. Describe how undertaking the aforementioned questions by this subproject furnishes synergic effect to the main project.

5. 執行方式/Methods and Approach

This section justifies the methods, technical choices, and the approaches in which solutions will be brought. A brief background information might be necessary. In addition, comment on how the past achievement of the co-PI(s) enables the success of the proposed approach.

Validate the diagnostic criteria for subtypes of PA.

To utilize the TAIPAI database to investigate the incidence of cancer in PA patients

We will categorize cancers into 17 groups and determine the incidence of each. The database will provide a unique opportunity to assess the impact of PA on long-term cancer risk, contributing significantly to our understanding of the malignant outcomes in PA patients.

To explore the clinical relevance of the aldosterone effect on Renal Cell Carcinoma (RCC).

In summary, this subproject will provide vital insights into the malignant outcomes in PA patients, validate diagnostic criteria for PA, and investigate the effects of target treatments on clinical outcomes. Importantly, we will also explore the clinical relevance of aldosterone's effect on RCC, providing valuable knowledge that can contribute to future clinical practice and treatment strategies for PA.

6. 本子計畫創新性與貢獻

Originality and Innovation of this Subproject and the Contribution to clinical problems

7. 參考文獻/References

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子計畫二 Subproject 2	子計畫名稱	醛固酮與腎細胞癌之細胞模型路徑和機制探討	主持人 PI	
	Title of Project	Aldosterone and Renal Cell Carcinoma: Elucidating Pathways and Mechanisms Across Model Systems		

繳交計畫前，請刪除說明/Please delete all the instructions before proposal submission.

1. 本子計畫目的/Specific Objectives of this Subproject

Describe, for this subproject, the specific objectives and the significance, e.g., prospects for academic excellence, originality and innovations in scientific research or technological development, and impact on the research field and international competitiveness.

This project is focused on outlining the specific objectives of our subproject, centered on exploring the role of aldosterone in renal cell carcinoma (RCC) signal transmission, its impacts on RCC cells, and the potential therapeutic effects of mTOR inhibitors and mineralocorticoid receptor (MR) antagonists on RCC.

Specific Aims 2.1: Decipher the Role of Aldosterone in RCC Signal Transmission

Specific Aims 2.2: Assess the Impact of Aldosterone on RCC Cells and the Underlying Mechanisms

Specific Aims 2.3: Evaluate the Therapeutic Potential of mTOR Inhibitor and MR Antagonist on RCC

In conclusion, this subproject aims to provide a detailed exploration of the role and impact of aldosterone on RCC and the potential therapeutic potential of mTOR inhibitors and MR antagonists. The findings could pave the way for effective therapeutic strategies to combat RCC in the future.

2. 經費編列

項目名稱	使用目的	總價
人事費	兼任研究助理(碩士級) (10000 元/月*12 月*1 人)	120,000
		小計： 元
業務耗材費		
消耗性試劑耗材	細胞株(Human RCC cell lines 786-O and A-498)及細胞培養相關試劑 (包含培養基，生長因子，血清，抗生素)	80,000
消耗性試劑耗材	離心管 (50cc/15cc/2cc/1.5cc); 分裝滴管 / 尖 (10mL/1mL/200μL/10μL); 細胞培養盤 (10cm Dish, 6well, 96well Plate) Transwell for migration assay; 相關抗體(pAKT, AKT, pmTOR, mTOR, Caspase 3, p53, Bax, BCL2, HIF1-a, S6, pS6, surviving, PCNA, Actin and GAPDH etc.); 小分子化合物、重組蛋白/化學藥品、緩衝溶液; 蛋白質分	700,000

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	離凝膠/轉漬試劑與耗材	
	論文投稿費及英修審查用	100,000
		小計： 元

總計：新台幣 100 萬 元

3. 人力配置

類別	姓名	現職	在本計畫內擔任之工作性質、項目及範圍
子計畫主持人	闕士傑	台大醫院泌尿部主任	計畫執行，監測主導實驗設計，實驗結果分析與討論，計畫報告之撰寫
協同主持人			協助計畫執行，協助實驗設計，協助實驗結果分析與討論，協助計畫報告之撰寫
研究人員			實驗室作業、耗材 管理、行政處理、經費報支、實驗結果彙整

4. 本子計畫對整體計畫的貢獻/Contribution of this Subproject to the Synergic Integration

This section re-defines the questions of the main project yet from the aspect of this subproject. Describe how undertaking the aforementioned questions by this subproject furnishes synergic effect to the main project.

5. 執行方式/Methods and Approach

This section justifies the methods, technical choices, and the approaches in which solutions will be brought. A brief background information might be necessary. In addition, comment on how the past achievement of the co-PI(s) enables the success of the proposed approach.

6. 本子計畫創新性與貢獻

Originality and Innovation of this Subproject and the Contribution to clinical problems

7. 參考文獻/References

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子計畫三 Subproject 3	子計畫名稱	醛固酮與轉移性腎細胞癌之關聯性的外泌體:多體學探勘	主持人 PI	黃俊仁
	Title of Project	Exosomes associated with aldosterone and metastatic renal cell carcinoma: multi-Omics approach		

1. 本子計畫目的/Specific Objectives of this Subproject

With an integrated Omics study, we aim to discover biomarkers to achieve early diagnosis and real-time disease monitoring. Clinical populations will focus on RCC patients with or without primary aldosteronism. We also aim to achieve differential diagnosis and strengthen the understanding of pathological mechanisms. This can be fulfilled by integrating novel biomarkers distinct among RCC patients with or without primary aldosteronism. Finally, the integrated Omics analysis combined with cells, animal models and clinical patient samples is used to find the pathogenic mechanism and provide clinicians with the basis for drug use. Specific aims and collaborations with other subprojects are summarized as follows.

Aim 1: To discover early diagnosis and prognosis biomarkers from clinical samples, including RCC patient with PA or without PA. Clinical samples including blood, urine and tissue will be collected and analyzed. The sample collection and exosome purification, characterization is collaborated with subproject 1, and the omics results will also be introduced to subproject 1 for correlation analysis. The finding is significant in early diagnosis (prediction) and is essential in selecting accurate clinical therapy (Collaborate with subproject 1 and 2).

Aim 2: To discover aldosterone infusion accelerates primary tumor size and metastatic spread of renal cancer cells from animal model. (Collaborate with subproject 2). The purified exosomes from cell or animal model set up in subproject 2 will be analyzed with proteomics and metabolomics approaches. The results will be transferred to subproject 1 for correlation analysis to reveal the most significant canonical pathways for aldosterone in the pathogenesis and progression of renal cell carcinoma.

Aim 3: To observe the pathogenic mechanism for early diagnosis and monitoring of aldosterone in the pathogenesis and progression of renal cell carcinoma. To discovery the pathogenic mechanism of aldosterone in the pathogenesis and progression of renal cell carcinoma, we will use samples from different sources (from subproject 1 and subproject 2) and combine multi-Omics approach at different stages to find important evidence. Canonical pathway analysis, a bioinformatics approach, scrutinizes gene-protein interactions within precise biological processes, unveiling pivotal signaling pathways. Leveraging gene expression data and protein networks, this method unveils regulatory mechanisms, offering insights for therapeutic targets and disease management.

In conclusion, this study will offer valuable insights into the molecular mechanisms underlying aldosterone-induced tumor growth and metastasis in RCC. It will also provide a rich resource of potential molecular targets for future therapeutic interventions. These aims align with our overarching goal of unraveling the genetic landscape in metastatic RCC via a transcriptomic approach.

2. 經費編列

項目名稱	使用目的	總價
人事費	碩士級助理協助執行本計畫相關實驗	474,340
		小計： 474,340 元
業務耗材費	外泌體分離純化與分析	175,660
	蛋白質體學與轉錄體學分析	450,000
		小計： 525,660 元

總計：新台幣 1,000,000 元

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3. 人力配置

類別	姓名	現職	在本計畫內擔任之工作性質、項目及範圍
子計畫主持人	黃俊仁	萬芳副院長	研究及策劃本項計畫之執行並督導進度之達成
協同主持人	張肇源	萬芳研究員	協助計畫執行，協助實驗設計，協助實驗結果分析與討論，協助計畫報告之撰寫
研究人員	蔡宜珍	碩士研究助理	協助執行本計畫相關實驗

4. 本子計畫對整體計畫的貢獻/Contribution of this Subproject to the Synergic Integration

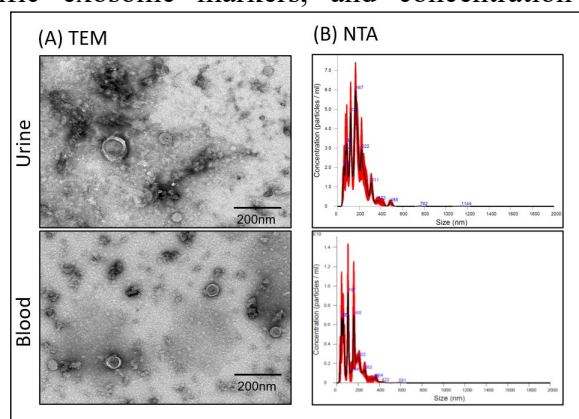
The integrated project aims to explore aldosterone's role in renal cell carcinoma pathogenesis. Subproject 3 will conduct omics analyses (RNA, proteins, metabolites) with a discovery phase followed by targeted high throughput analysis. Biofluids from subprojects 1 and 2, including exosomes isolated via ultracentrifugation, will be used. Biomolecules and pathways identified will integrate with biochemical analysis and clinical data from subprojects 1 and 2. Subproject 3 will also validate analytical methods for future practical use.

5. 執行方式/Methods and Approach

5.1 Collection and usage of samples from cell model, animal model and clinical cohort

Exosomes from cell model, animal model and clinical patients, including medium and liquid biopsy (urine and plasma), will be collected from the subproject 1 and 2. Purification will be conducted by ultracentrifugation. The conformation, size, specific exosome markers, and concentration of collected exosomes will be characterized.

Our preliminary data demonstrate that exosomes can be isolated from liquid biopsies. The QC passed samples will be transferred for proteomics and metabolomics analysis. Cells, animal samples, and clinical patient samples will be collected. All samples will be extracted by RIPA buffer. The same procedure will be done for Omics analysis. The results will be used to find the potential pathways related to RCC with or without PA early diagnosis and progression.



5.2 Omics analysis

Proteomic and miRNA analyses were conducted using both LC-MS/MS and small RNA sequencing methods. The relevant specimens were derived from extracellular vesicles in urine and blood, as well as tissue samples from the kidneys. Proteomic and miRNA analyses were performed following the recommended protocols provided by the manufacturers, and subsequent analysis was carried out after quality control (QC) procedures.

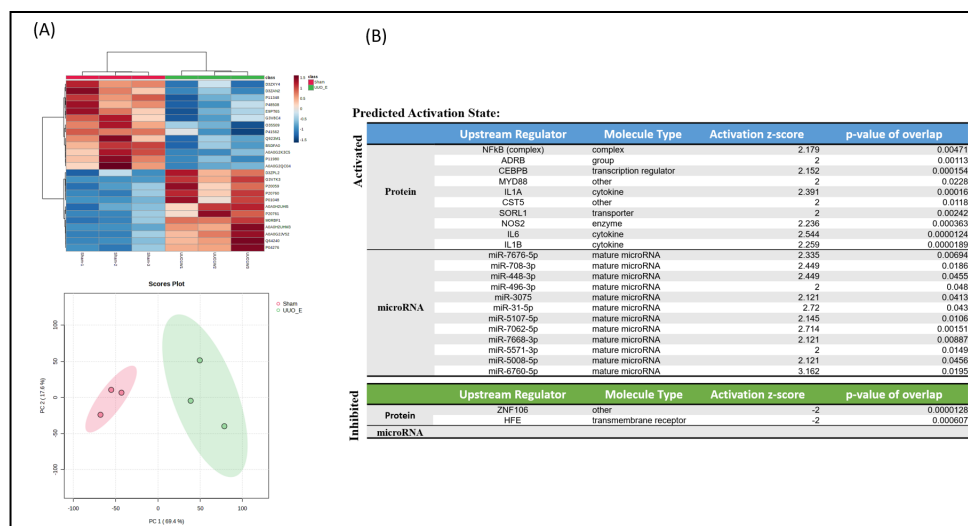
5.3 Data analysis and interpretation

In our pursuit of a comprehensive understanding, the data we have amassed will undergo a rigorous examination alongside the Go and KEGG databases. This endeavour will be facilitated by the utilization of a sophisticated biological information database known as the Ingenuity Pathway Analysis (IPA). The overarching goal of this analysis is to discern significant protein biomarkers or mRNA biomarkers that hold potential relevance within the context of our research.

Our preliminary data demonstrate that a comparative proteomic analysis of urinary exosomes between the one-week postoperative group and the normal group following early unilateral ureteral obstruction surgery and potential biomarkers of activation and inhibition in predicted pathways of early subacute renal disease by IPA analysis.

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6. 本子計畫創新性與貢獻

This integrated project delves into the intricate role of aldosterone in the genesis and progression of renal cell carcinoma. Utilizing the diverse analytical capabilities encompassing proteomics, transcriptomics, and other multi-dimensional assessments offered by sub-project 3, our primary objective is to elucidate the underlying pathogenic mechanisms through which aldosterone contributes to the development and tumorigenesis of renal cell carcinoma. Simultaneously, we endeavor to identify potential biomarkers for early-stage diagnosis, encompassing varying stages of the condition. Moreover, by amalgamating the insights derived from cellular, animal model, and clinical patient data, our approach seeks to unravel the intricate pathogenic pathways while identifying potential avenues for therapeutic intervention. Our overarching aim is to provide clinicians with an informed foundation for therapeutic decision-making, thereby enhancing the prospects for efficacious medical intervention.

7. 參考文獻/References

1. Lu Y, Zhang M, Zhou J, Liu X, Wang L, Hu X, Mao Y, Gan R, Chen Z. **Extracellular vesicles in renal cell carcinoma: challenges and opportunities coexist**. Front Immunol. 2023, 14:1212101.
2. Xu J, Hu Z, Cao H, Zhang H, Luo P, Zhang J, Wang X, Cheng Q, Li J. **Multi-omics pan-cancer study of cuproptosis core gene FDX1 and its role in kidney renal clear cell carcinoma**. Front Immunol. 2022, 20:13:981764.
3. Zhao J, Liang J, Yang Y, Sun G, Zhang X, Zhao J, Hu X, Chen J, Zhu S, Ni Y, Zhang Y, Dai J, Wang *et al*: **Integrated multi-omics analyses reveal that BCAM is associated with epigenetic modification and tumor microenvironment subtypes of clear cell renal cell carcinoma**. Clin Epigenetics. 2022, 14(1):99.

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四、個人資料表及績效表/Curriculum Vitae of Principal Investigators and Major Research Accomplishments in the Recent 5 Years

參與計畫的 Co-PI 皆須提供以下兩項資料：

1. 個人資料表：自科技部下載的個人資料表 C301 電子檔

CV: The submitted CV can be downloaded from your account at MoST.

2. 五年內研究表現：請提供科技部各學術司專屬表格

Major Research Accomplishments in the Recent 5 Years: The form can be downloaded from the following link

科技部各學術司之「專屬表格」名稱

人文司 科技部人文司專題研究計畫主持人代表性研究成果表

自然司 自然司專題計畫主持人近五年研究成果

自然司防災科技學門及永續科學學門專題計畫主持人近五年研究成果

工程司 科技部工程司專題計畫主持人近五年成果績效表

生科司 科技部生命科學司學術研究績效表

科國司 科技部科教國合司研究人員近五年研究成果及貢獻調查表

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個人資料表

基本資料			填表日期：____/____/____		
中文姓名		英文姓名	(Last Name) (First Name) (Middle Name)		
國籍		性別	☐ 男 ☐ 女	出生日期	19____年____月____日
聯絡地址					
聯絡電話	(公) (宅 / 手機)				
傳真號碼		E-mail			
主要學歷 (由最高學歷依次填寫，若仍在學者，請在學位欄填「肄業」)					
學校名稱	國別	主修學門系所	學位	起訖年月(西元年/月)	
				自 ____/____/____ 至 ____/____/____	
				自 ____/____/____ 至 ____/____/____	
				自 ____/____/____ 至 ____/____/____	
				自 ____/____/____ 至 ____/____/____	
現職及與專長相關之經歷 (指與研究相關之專任職務，請依任職之時間先後順序由最近者往前追溯)					
服務機構	服務部門／系所		職稱	起訖年月(西元年/月)	
現職：				自 ____/____/____ 至 ____/____/____	
經歷：				自 ____/____/____ 至 ____/____/____	
				自 ____/____/____ 至 ____/____/____	
				自 ____/____/____ 至 ____/____/____	
專長 (請自行填寫與研究方向有關之學門及次領域名稱)					
1.	2.	3.	4.		

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科技部生科司學術研究績效表

2019/10/29

姓名： 職稱： 服務機關系所：

*若為新進學者，是否曾參加本司舉辦之「生命科學新進教師學者研習營」？

☐是，參加年度：_____年，☐否。

一、請簡述代表性研究成果或實作之重要貢獻(至多1頁)

二、近五年內之研究成果，其中最具代表性研究成果論文(5篇為限)

1. 請依發表時間之先後順序填寫，學術論文必須填寫所有作者(按原出版之次序，通訊作者請加註*)、出版年、月份、題目、期刊名稱、卷期及起迄頁數等資訊。
2. 五年內曾生產或請育嬰假者，得依胎次再延長，每胎次延長二年，曾服國民義務役者，得依實際服役時間予以延長，但應檢附相關證明文件。

序 號	論文資料	*必填 1.突破性之創見 2.對學術發展、社會、經濟等面向之影響
範 例	Hsu, W.-H., Wang, W.-J., Lin, W.-Y., Juang, Y.-M., Lai, C.-C., Liao, J.-C., and Chen, H.-C.* (2018, Jun). Adducin-1 is essential for spindle pole integrity through its interaction with TPX2.. EMBO Reports , e45607.	XXXXXXXXXXXXXXXXXXXXXXXXXXXX.....
1		
2		
3		
4		
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三、近五年內獲獎情形及重要會議邀請演講(至多5項)